Free-body diagrams, part 2

LEARNING TARGET · R9 I can find net force, then acceleration, straight from the FBD — inclines included.

A · KEY VOCABULARY

Normal force	Points perpendicular to (straight away from) the surface — on a ramp, it is tilted.
Weight	Always points straight down, toward Earth.
Component	The downhill part of weight. A way to calculate — not an extra force.
Accel. direction	Same direction as the net force.

B · KEY EQUATIONS + UNITS

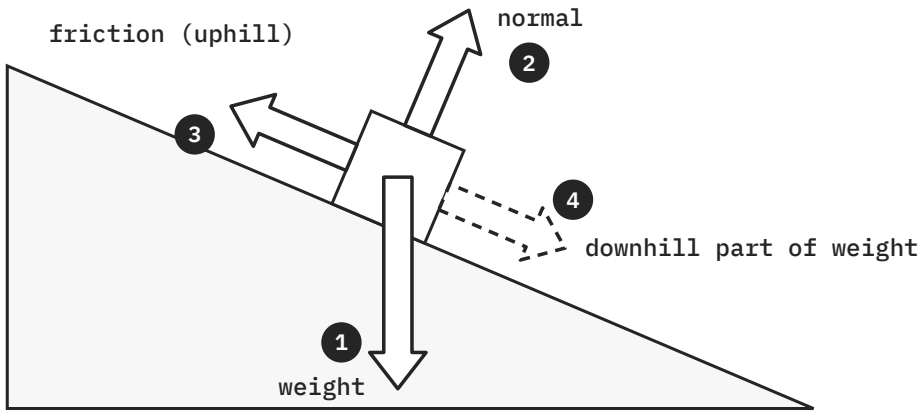
$$\Sigma F_x = m\,a_x \cdot \Sigma F_y = m\,a_y$$

down ramp: $mg \sin\theta$
into ramp: $mg \cos\theta$

+x down the ramp
+y out of the surface

C · COLOR IT · FORCES ON A RAMP

Don't double count weight.



- ☐ **1 · ORANGE** Weight (W)
- ☐ **2 · ORANGE** Normal force (N)
- ☐ **3 · ORANGE** Friction (f)
- ☐ **4 · ORANGE** Downhill part (dashed)

D · WORKED EXAMPLE

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration.

GIVEN	EQUATION	WORK	ANSWER
m = 2 kg 7 N downhill 3 N uphill	$a = F_{net} \div m$	$F_{net} = 7 - 3 = 4$ N downhill $a = 4 \div 2$	**2 m/s² downhill**

E · THE 2026-XJ THREAD

DAY 16 OF 22 · NEW TOOL: FBD → ACCELERATION

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else